Predicting Falcon 9 First-Stage Landings

A data-driven study of launch history, landing outcomes, interactive analytics and classification.

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GitHub repository: ADD YOUR REPOSITORY URL

Executive Summary

Goal: predict whether the Falcon 9 first stage lands successfully.

Methods: SpaceX API, Wikipedia web scraping, data wrangling, SQL, EDA, Folium, Plotly Dash and classification.

Observed ML result: Decision Tree achieved 88.89% test accuracy on an 18-record holdout (16 correct).

Caution: small test set and preprocessing leakage in the course workflow limit generalisation.

Introduction and Research Question

Reusable boosters can lower launch costs; successful recovery is therefore an operationally meaningful outcome.

Question: which launch, vehicle and mission characteristics are associated with landing success?

Target variable: Class = 1 for successful landing; Class = 0 otherwise.

Analysis uses the supplied historical course datasets, not current live SpaceX launches.

Data Collection — SpaceX API

Source notebook: jupyter-labs-spacex-data-collection-api.ipynb

Request SpaceX launch records from the public API.

Join related rocket, payload, launchpad and core metadata using identifiers.

Convert nested JSON into a tabular launch-level dataset.

Review missing values and retain the relevant Falcon 9 records.

GitHub file URL: ADD LINK TO API NOTEBOOK

Data Collection — Web Scraping

Source notebook: jupyter-labs-webscraping.ipynb

Retrieve the historical Falcon 9 launch table from Wikipedia.

Parse HTML tables using BeautifulSoup / pandas and standardise field names.

Extract launch date, mission, payload, orbit and landing-related fields.

Use the scraped dataset for complementary historical exploration.

GitHub file URL: ADD LINK TO WEB SCRAPING NOTEBOOK

Data Wrangling Methodology

Source notebook: labs-jupyter-spacex-Data wrangling.ipynb

Clean and align launch records; handle missing or ambiguous landing outcomes.

Encode landing outcome as the binary Class label.

Construct a modelling feature table, including orbit, launch site and booster indicators.

Supplied dataset_part_2.csv: 90 rows; dataset_part_3.csv: 90 rows and 83 features.

GitHub file URL: ADD LINK TO DATA WRANGLING NOTEBOOK

Exploratory Data Analysis — Methods

Source notebook: edadataviz.ipynb

Explore relationships among launch site, payload mass, orbit, flight number and landing success.

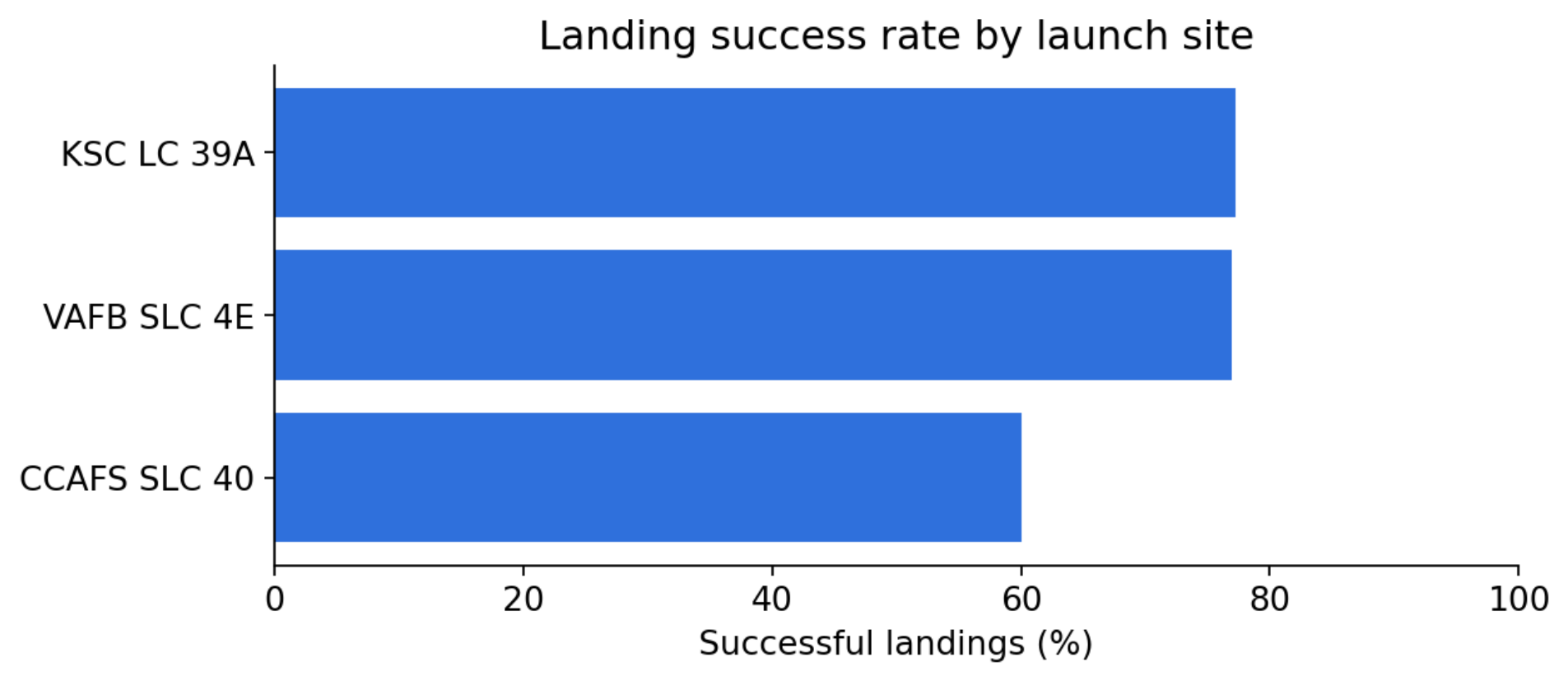
Use bar charts for categorical distributions, scatter plots for numeric associations, and line charts for time trends.

Interpret trends descriptively; they are not causal effects.

GitHub file URL: ADD LINK TO EDA NOTEBOOK

EDA Result — Launch Site Success

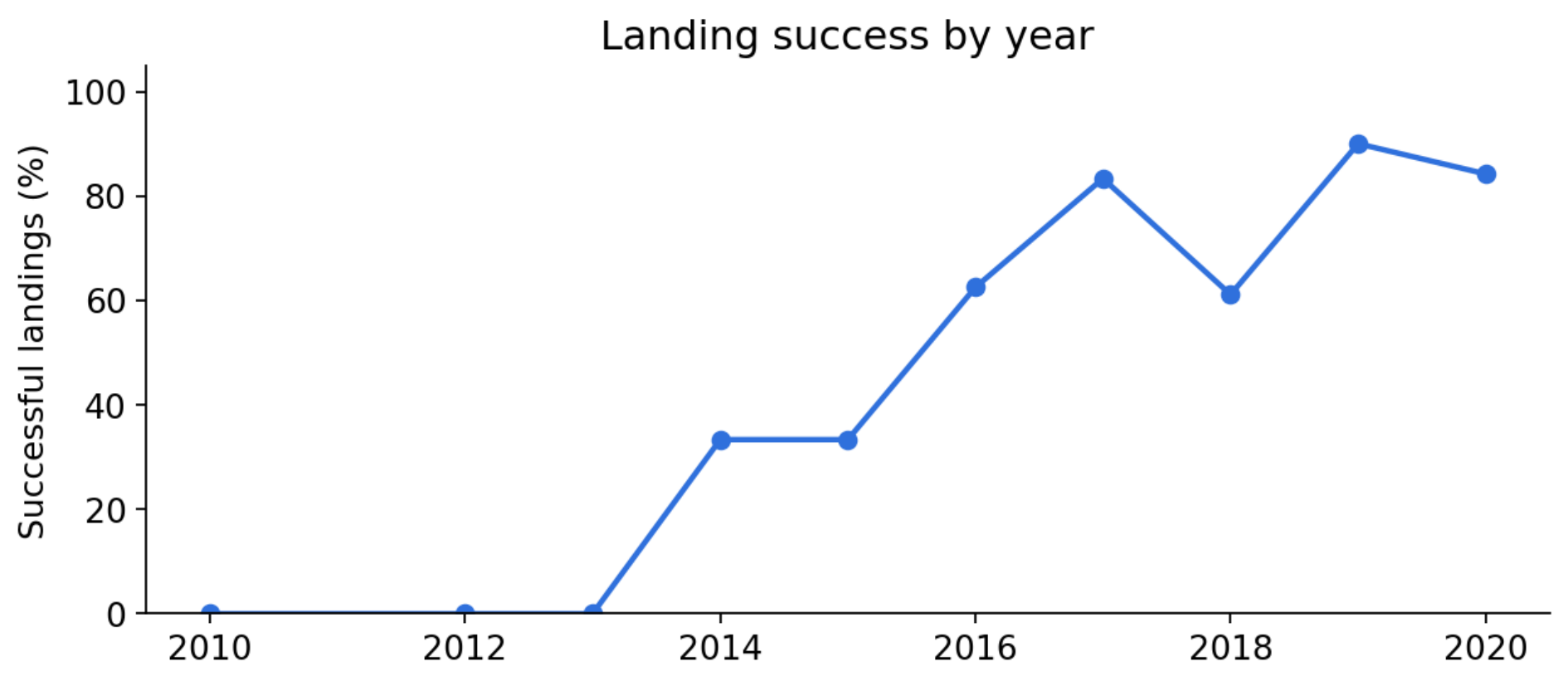
Calculated from the supplied dataset_part_2.csv (90 launches)



Bars show observed success share within each site; site sample sizes differ.

EDA Result — Yearly Landing Trends

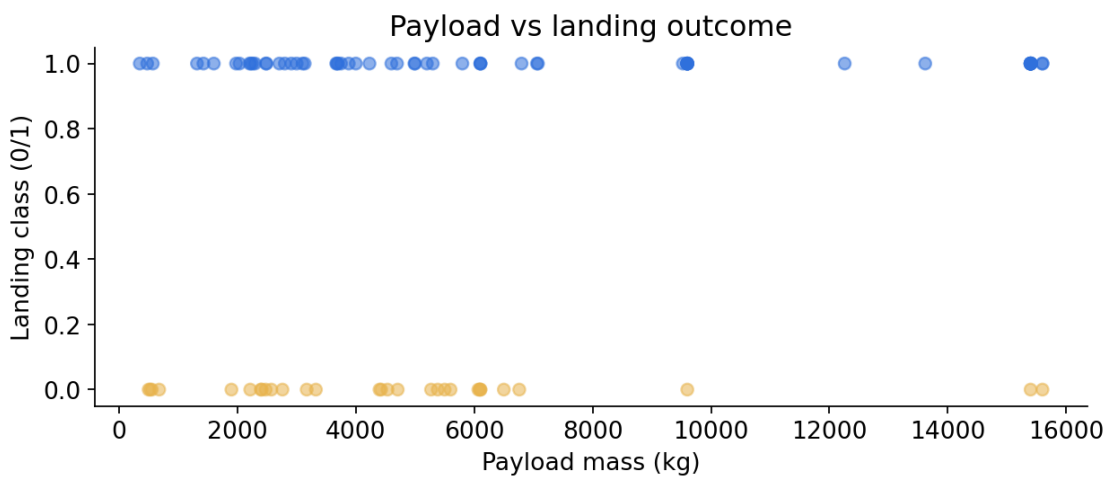
Calculated from launch dates and Class in dataset_part_2.csv

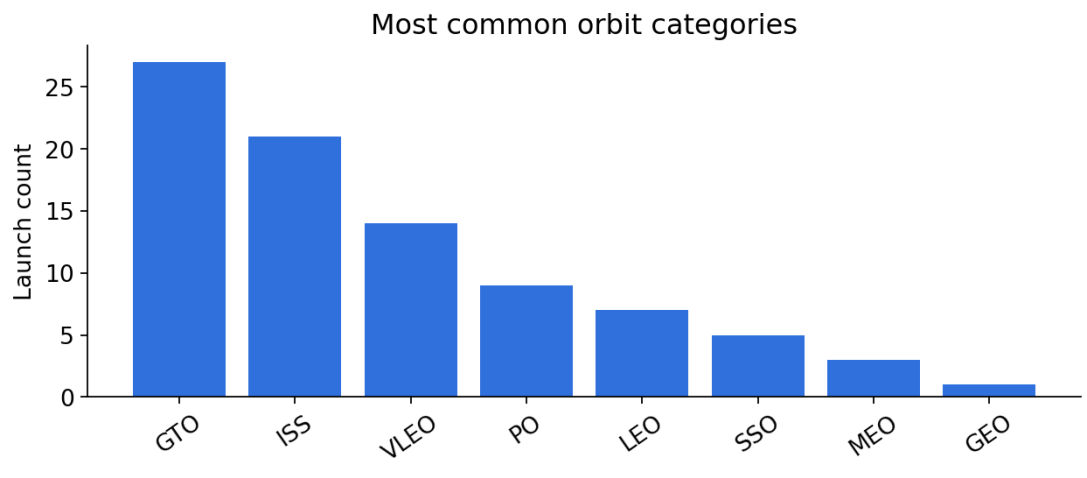


Descriptive yearly rates; early years have small sample sizes.

EDA Result — Payload and Orbit

Calculated from dataset_part_2.csv





EDA with SQL — Methodology

Source notebook: jupyter-labs-eda-sql-coursera_sqllite.ipynb

Load launch records into a relational table and use SELECT, WHERE, GROUP BY and ORDER BY.

Questions include launch site counts, payload extrema, mission categories and outcome frequencies.

Group by date or site to explore changes in activity and success over time.

SQL execution outputs have not been independently verified in this deliverable.

GitHub file URL: ADD LINK TO SQL NOTEBOOK

EDA with SQL — Results to Verify

Launch sites: aggregate launch counts and landing outcomes by site.

Payloads: inspect minimum, maximum and selected mission payload masses.

Success and rankings: group outcome counts and rank categories using SQL.

Time analysis: filter and aggregate by launch year.

ACTION: replace this slide with screenshots/tables from a successfully executed SQL notebook.

Interactive Visual Analytics

Source files: lab_jupyter_launch_site_location.ipynb and spacex-dash-app.py

Folium: map launch site coordinates and annotate markers, launches and nearby infrastructure.

Plotly Dash: launch-site dropdown, success pie chart, payload range slider and outcome scatter plot.

These interactive views complement static exploratory charts.

GitHub Folium URL: ADD LINK

GitHub Dash URL: ADD LINK

Folium Maps — Evidence Required

Map 1: launch-site markers and geographic labels.

Map 2: launch records / outcome markers where implemented.

Map 3: proximity analysis to coastlines, railways, highways or nearby infrastructure.

ACTION: insert genuine screenshots from the executed Folium notebook before submission.

Do not treat this slide as evidence that the maps have been rendered.

Plotly Dash — Evidence Required

Screenshot 1: launch-site dropdown and corresponding success pie chart.

Screenshot 2: payload slider and success-by-payload scatter plot.

Confirm callbacks update the visualisations when inputs change.

ACTION: insert genuine screenshots from the running dashboard before submission.

Dashboard source: spacex-dash-app.py

Predictive Analysis — Model Comparison

Observed results from the previously executed ML notebook



CV: 10-fold GridSearchCV best score; test set: 18 observations.

Predictive Analysis — Interpretation

Decision Tree achieved the highest observed test accuracy: 88.89% (16/18).

Logistic Regression, SVM and KNN each achieved 83.33% (15/18).

A confusion matrix is required for the final rubric: insert the verified plot from the executed notebook.

Important limitation: scaling before the train/test split creates data leakage; rerun with a pipeline for robust claims.

Conclusions and Next Steps

The project combines data acquisition, descriptive analytics, interactive visualisation and predictive modelling.

Observed landing outcomes vary across launch sites and time; these associations warrant further investigation.

Decision Tree led in the reported holdout comparison, but the small sample and leakage prevent a strong superiority claim.

Next: leakage-free pipelines, repeated cross-validation, calibration and evaluation on newer launches.

Repository and Submission Checklist

Repository: ADD YOUR PUBLIC GITHUB URL

Upload all seven notebooks, Dash app, supporting CSV files, README and this PDF.

Replace placeholder GitHub links with direct file links.

Replace SQL, Folium and Dash evidence slides with real results; add confusion matrix.

Verify all links and export this PDF as: Data Science Capstone Project Report.pdf